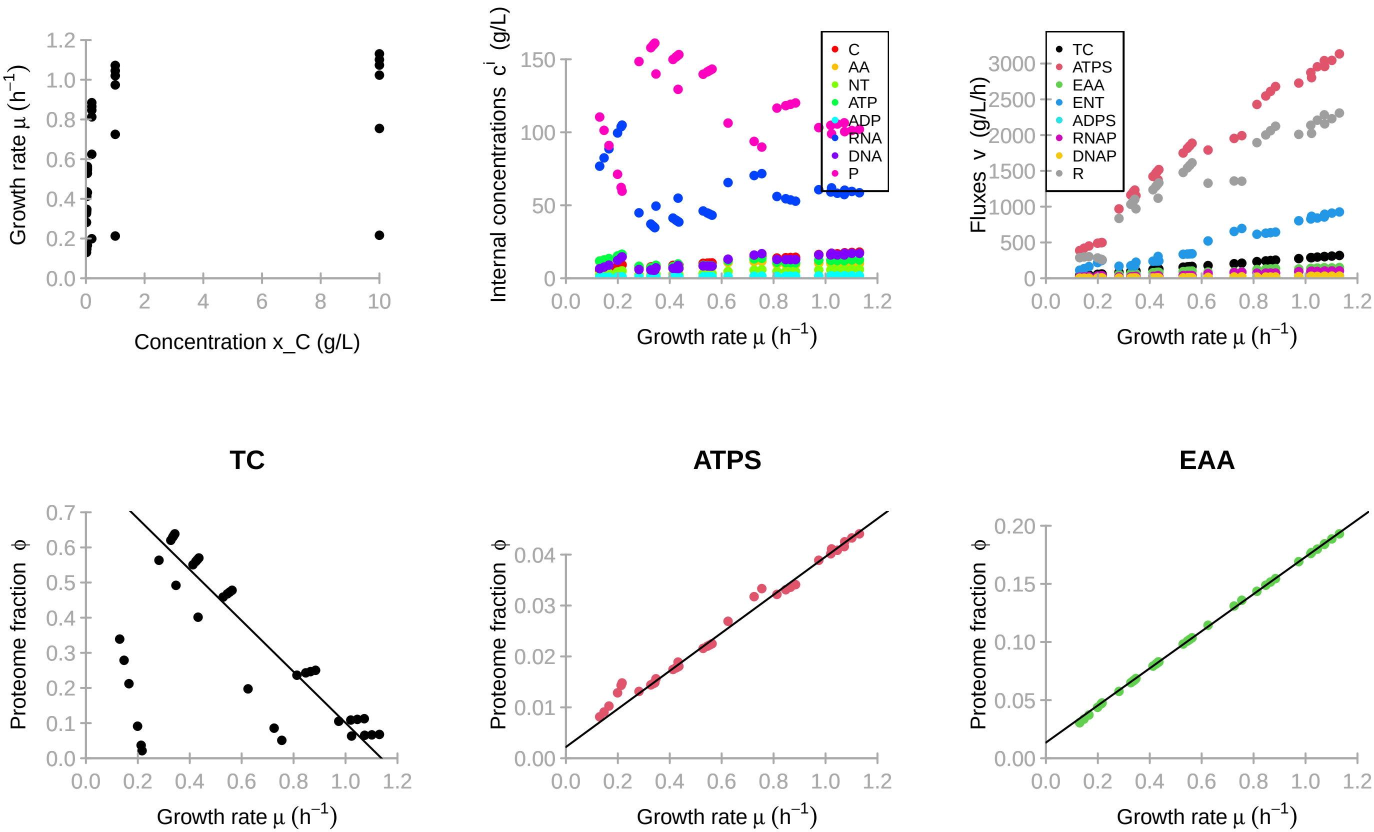
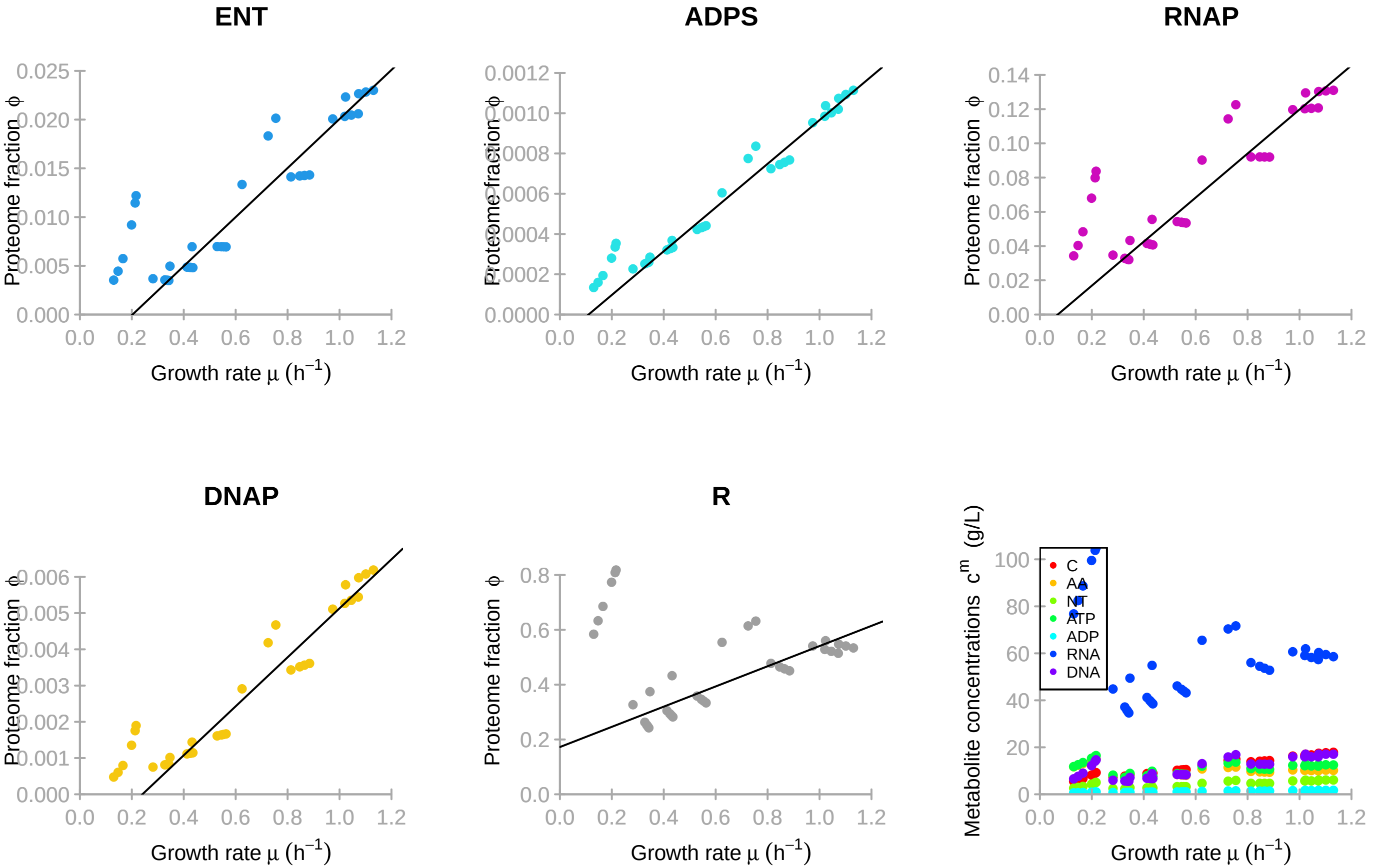
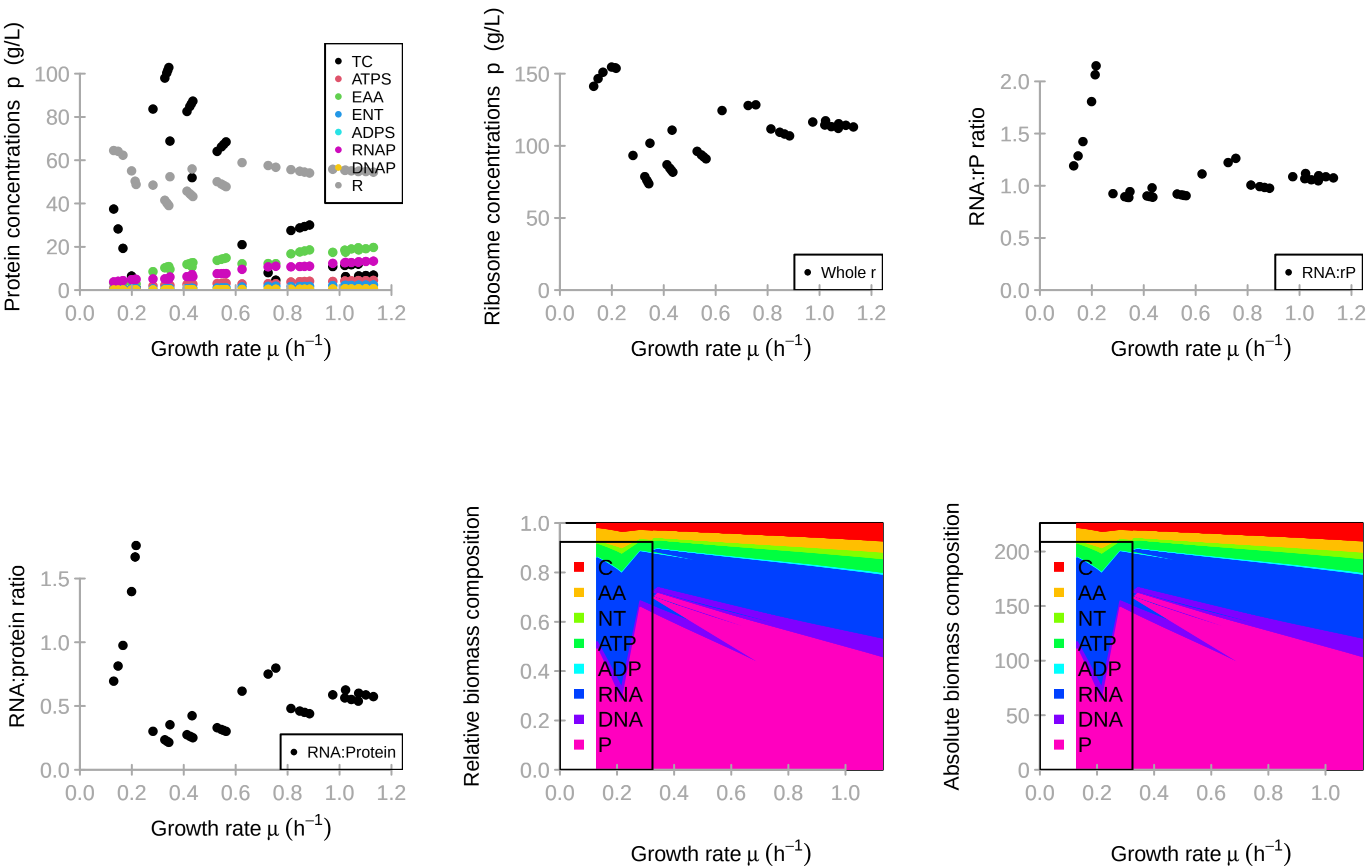
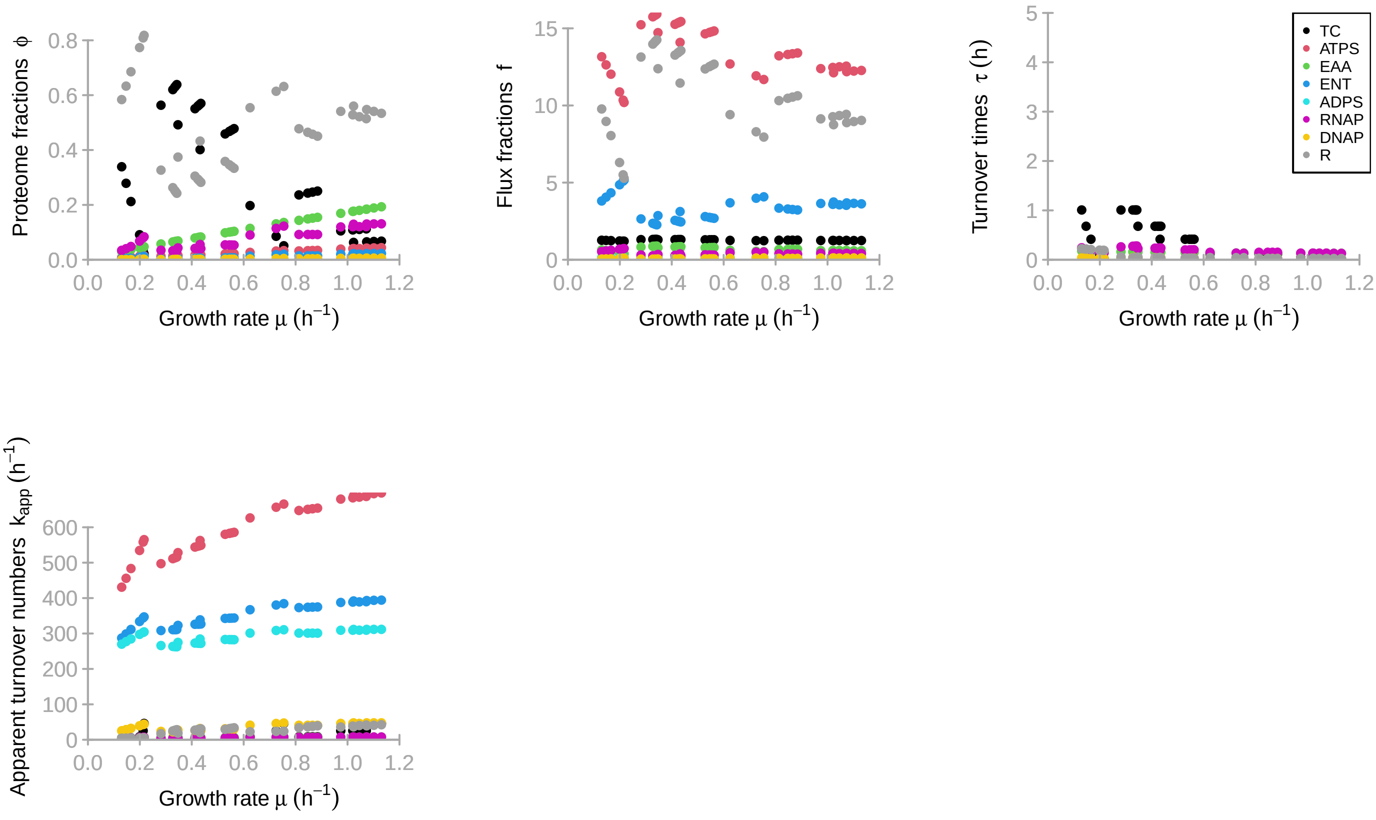

Growth rate μ (h−1)
Concentration x_C (g/L)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
2
4
6
8
10
Internal concentrations ci (g/L)
0
50
100
150
C
AA
NT
ATP
ADP
RNA
DNA
P
Growth rate μ (h−1)
Fluxes v (g/L/h)
0
500
1000
1500
2000
2500
3000
TC
ATPS
EAA
ENT
ADPS
RNAP
DNAP
R
Growth rate μ (h−1)
TC
ATPS
EAA
Proteome fraction ϕ
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.00
0.01
0.02
0.03
0.04
0.00
0.05
0.10
0.15
0.20
Growth rate μ (h−1)
Growth rate μ (h−1)
Growth rate μ (h−1)


ENT
ADPS
RNAP
DNAP
R
Proteome fraction φ
Growth rate μ (h⁻¹)
Metabolite concentrations c^m (g/L)
C
AA
NT
ATP
ADP
RNA
DNA



Protein concentrations p (g/L)
TC
ATPS
EAA
ENT
ADPS
RNAP
DNAP
R
Growth rate μ (h^{-1})
Ribosome concentrations p (g/L)
Whole r
Growth rate μ (h^{-1})
RNA:rP ratio
RNA:rP
Growth rate μ (h^{-1})
RNA:protein ratio
RNA:Protein
Growth rate μ (h^{-1})
Relative biomass composition
C
AA
NT
ATP
ADP
RNA
DNA
P
Growth rate μ (h^{-1})
Absolute biomass composition
C
AA
NT
ATP
ADP
RNA
DNA
P
Growth rate μ (h^{-1})



Proteome fractions ϕ
Growth rate μ (h^{-1})
Flux fractions f
Growth rate μ (h^{-1})
Turnover times τ (h)
Growth rate μ (h^{-1})
TC
ATPS
EAA
ENT
ADPS
RNAP
DNAP
R
Apparent turnover numbers k_{app} (h^{-1})
Growth rate μ (h^{-1})

M

	TC	ATPS	EAA	ENT	ADPS	RNAP	DNAP	R
C	1	−0.17	−1	−0.04	0.13	0.36	0.36	0.16
AA	0	0	1	0	0	0	0	−0.06
NT	0	0	0	0.19	−1	−1	−1	0
ATP	0	0.98	0	−0.96	0	0	0	−0.94
ADP	0	−0.83	0	0.81	0.87	0	0	0.79
RNA	0	0	0	0	0	0.64	0	0
DNA	0	0	0	0	0	0	0.64	0
P	0	0	0	0	0	0	0	0.05

K

	TC	ATPS	EAA	ENT	ADPS	RNAP	DNAP	R
x_C	1	0	0	0	0	0	0	0
x_W	0	1	0	0	0	0	0	0
x_INH	0	0	0	0	0	0	0	0
C	10.6	3.5	3.5	3.5	10.6	0	0	0
AA	0	0	2.3	0	0	0	0	0.8
NT	0	0	0	2.3	0.8	0.8	0.8	0
ATP	0	2.3	0	0.8	0	0	0	0.8
ADP	0	0.2	0	0.7	0.7	0	0	0
RNA	0	0	0	0	0	0	0	0
DNA	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0

KA

	TC	ATPS	EAA	ENT	ADPS	RNAP	DNAP	R
x_C	0	0	0	0	0	0	0	0
x_W	0	0	0	0	0	0	0	0
x_INH	0	0	0	0	0	0	0	0
C	0	0	0	0	0	0	0	0
AA	0	0	0	0	0	0	0	0
NT	0	0	0	0	0	0	0	0
ATP	0	0	0	0	0	0	0	0
ADP	0	0	0	0	0	0	0	0
RNA	0	0	0	0	0	0	0	72
DNA	0	0	0	0	0	11	11	0
P	0	0	0	0	0	0	0	0

kcat

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]
kcatf	50.6	931.1	9.1	501.7	352.7	14.4	88.8	108.4
kcatb	0	0	0	0	0	0	0	0

Keq

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]
[1,]	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf